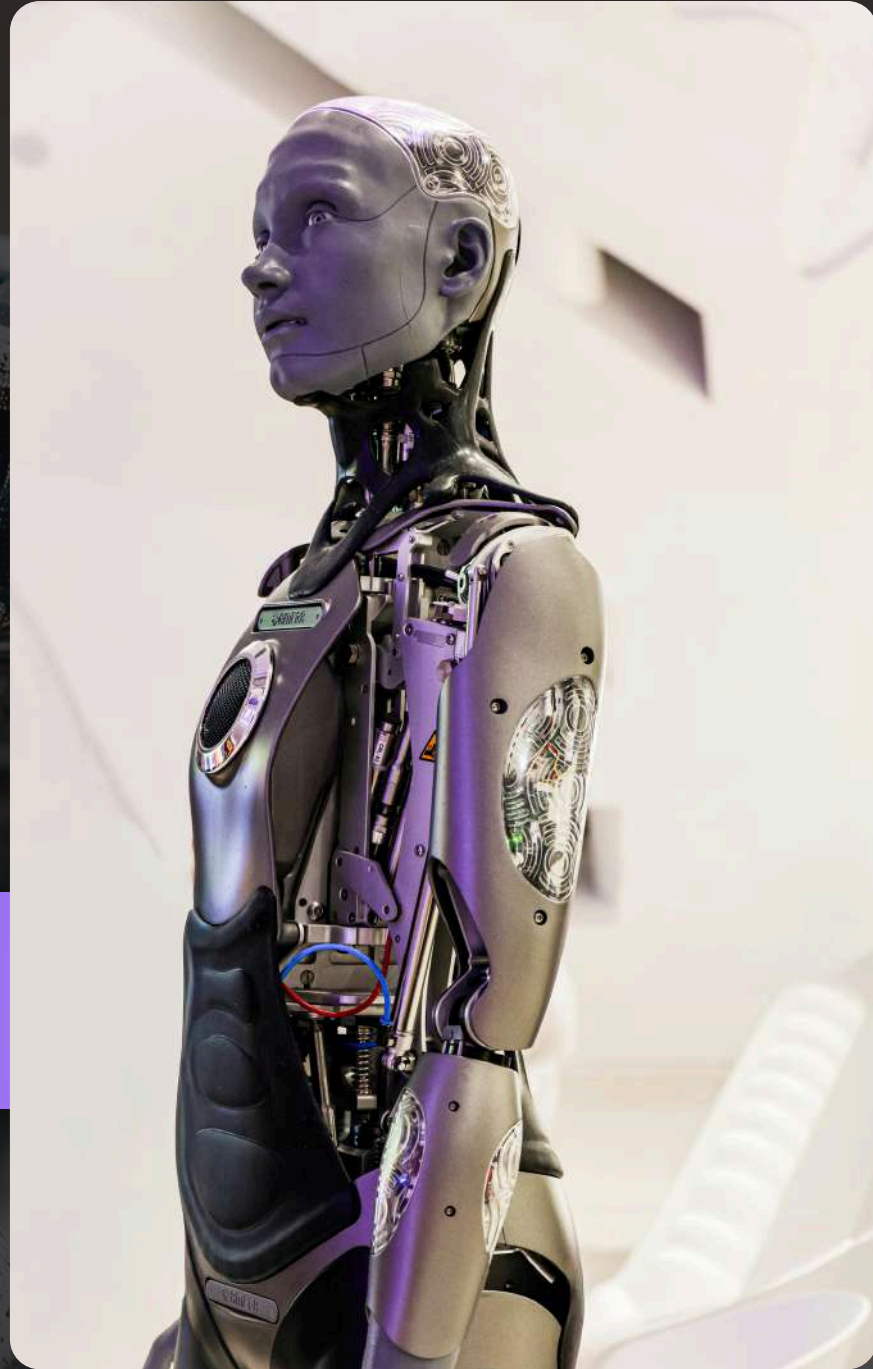
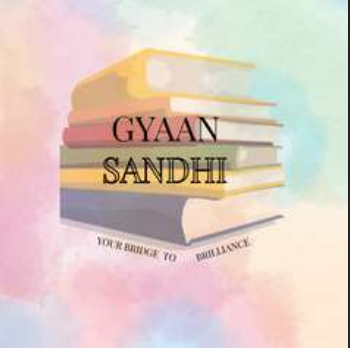




UNIT 5

Deep Learning

Deep Generative Models

Deep Generative Models

Introduction

- Deep Generative Models are a type of deep learning models.
- They learn to generate new data similar to the data used during training.
- Unlike traditional models that only predict labels, generative models learn the data distribution.
- These models can create new realistic examples such as images, text, audio, and videos.
- Example: A model trained on thousands of cat images can generate a completely new cat image that looks real.



Deep Generative Models

Features

- Learn hidden patterns and structures in data.
- Generate realistic and synthetic data.
- Work with images, text, speech, and music.
- Useful in creative AI applications.
- Improve data augmentation and simulation.

Applications



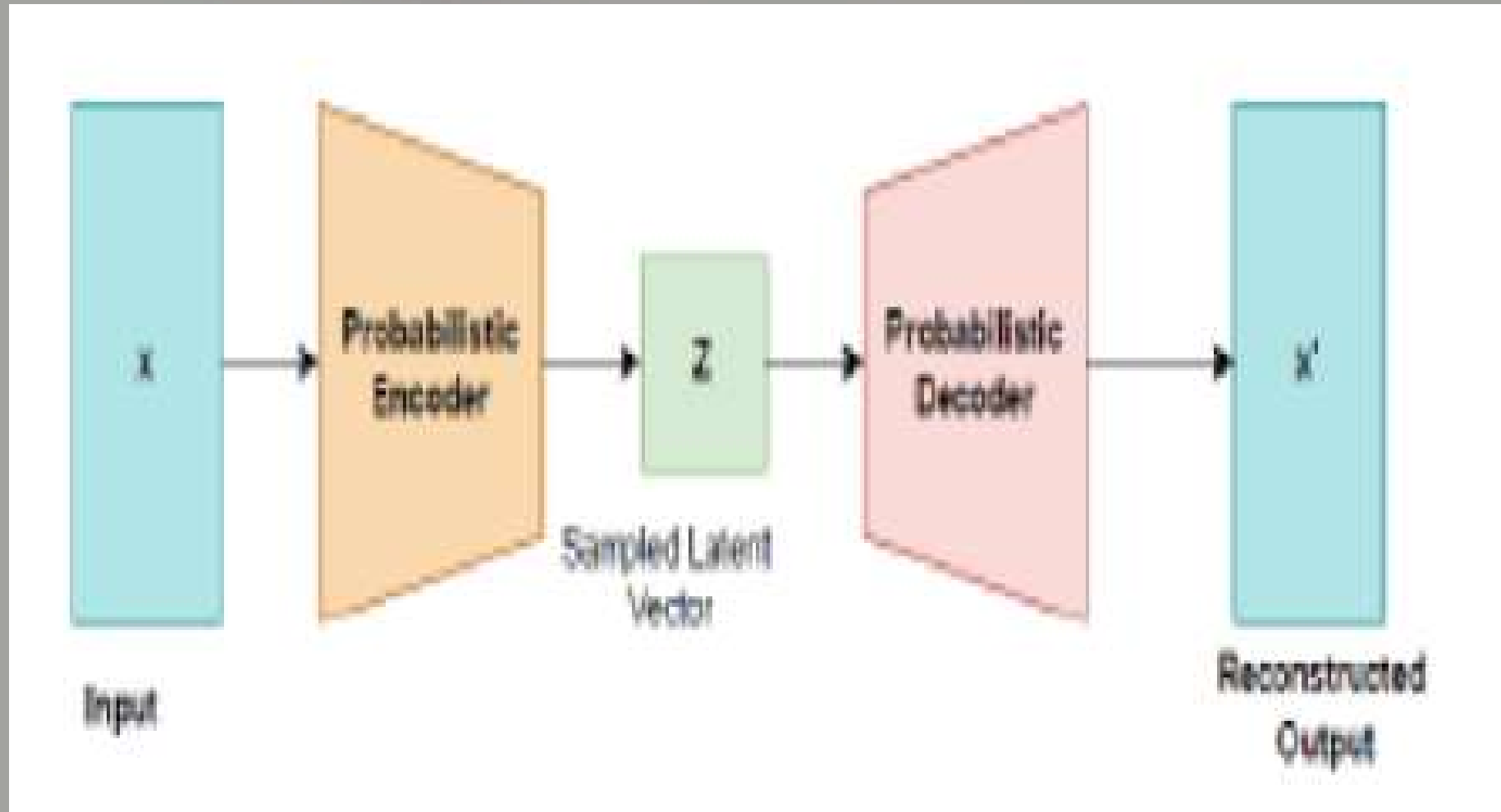
Deep Generative Models

- Computer Vision
- Natural Language Processing (NLP)
- Audio and Speech Generation
- Image Generation
- Video Creation
- Medical Imaging
- Chatbots and AI Assistants

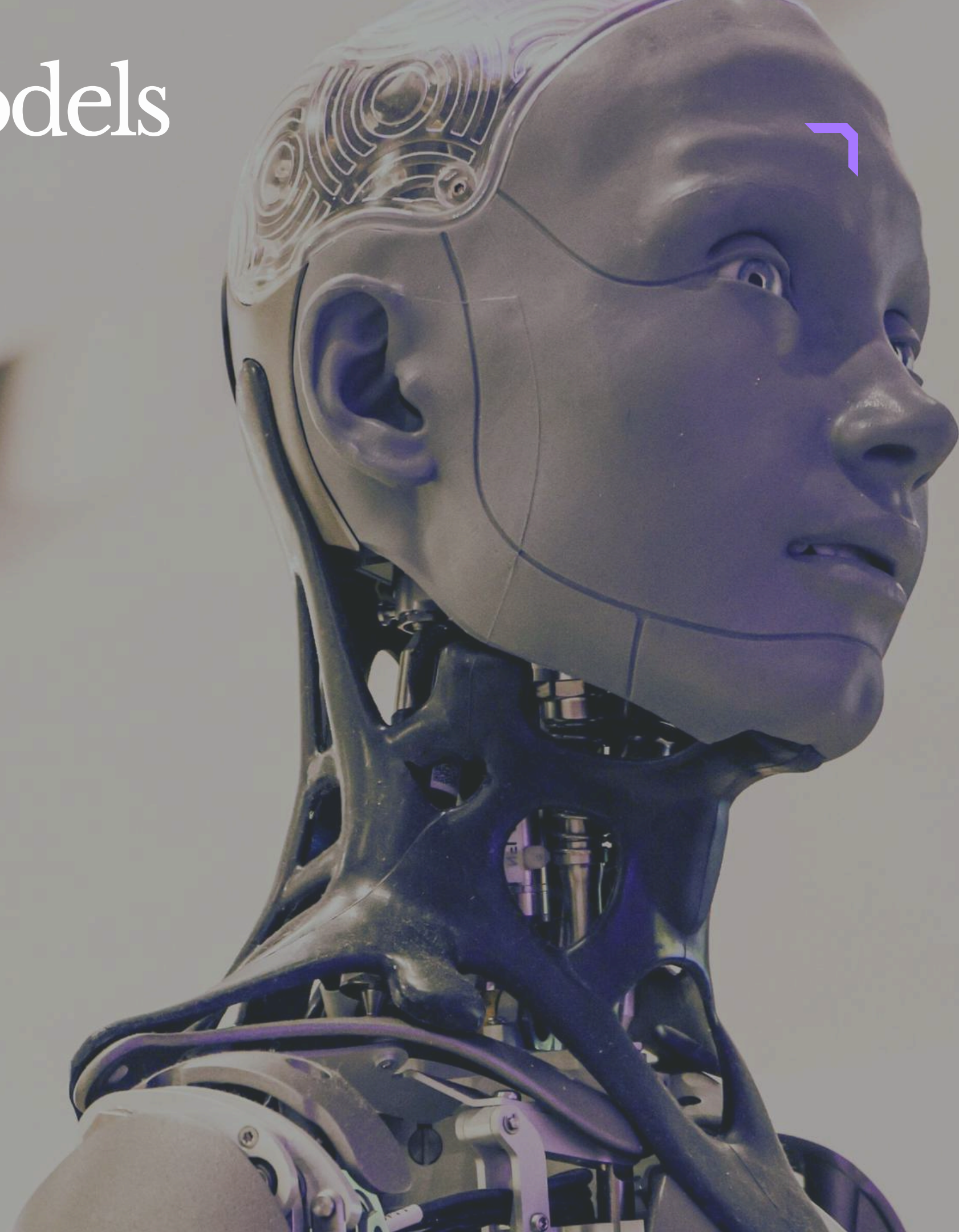
Diagram



Deep Generative Models



Types of Deep Generative Models



Deep Generative Models

Types of Deep Generative Models

1. Variational Autoencoders (VAEs)

Definition

- VAEs compress input data into a smaller latent representation (encoding).
- Then they reconstruct or generate similar data through decoding.

Working

- Encoder converts input into latent vector.
- Latent space stores compressed features.
- Decoder reconstructs output from latent vector.



Deep Generative Models

Types of Deep Generative Models

Advantages

- Smooth latent space representation.
- Good for image generation and compression.
- Stable training process.

Example

- Generating handwritten digits similar to MNIST dataset.

2. Generative Adversarial Networks (GANs)

Definition

- GAN consists of two neural networks working against each other.



Deep Generative Models

Types of Deep Generative Models

Components

a) Generator

- Creates fake or synthetic data.

b) Discriminator

- Detects whether data is real or fake.

Working

- Generator tries to fool the discriminator.
- Discriminator improves at detecting fake data.
- Both networks improve continuously.

Advantages

- Generates highly realistic images.
- Produces high-quality synthetic content.



Deep Generative Models

Types of Deep Generative Models

3. Autoregressive Models

Definition

- These models generate data step by step.
- Each next value depends on previous generated values.

Working

- Predict next word, pixel, or sound based on earlier outputs.

Advantages

- Good sequence modeling.
- Effective for text and speech generation.



Generative Adversarial Network (GAN)

Introduction

- Generative Adversarial Network (GAN) is a type of deep generative model.
- It is used to generate new and realistic data such as images, audio, videos, and text.
- GAN learns patterns from training data and creates similar synthetic data.
- Introduced by Ian Goodfellow in 2014.

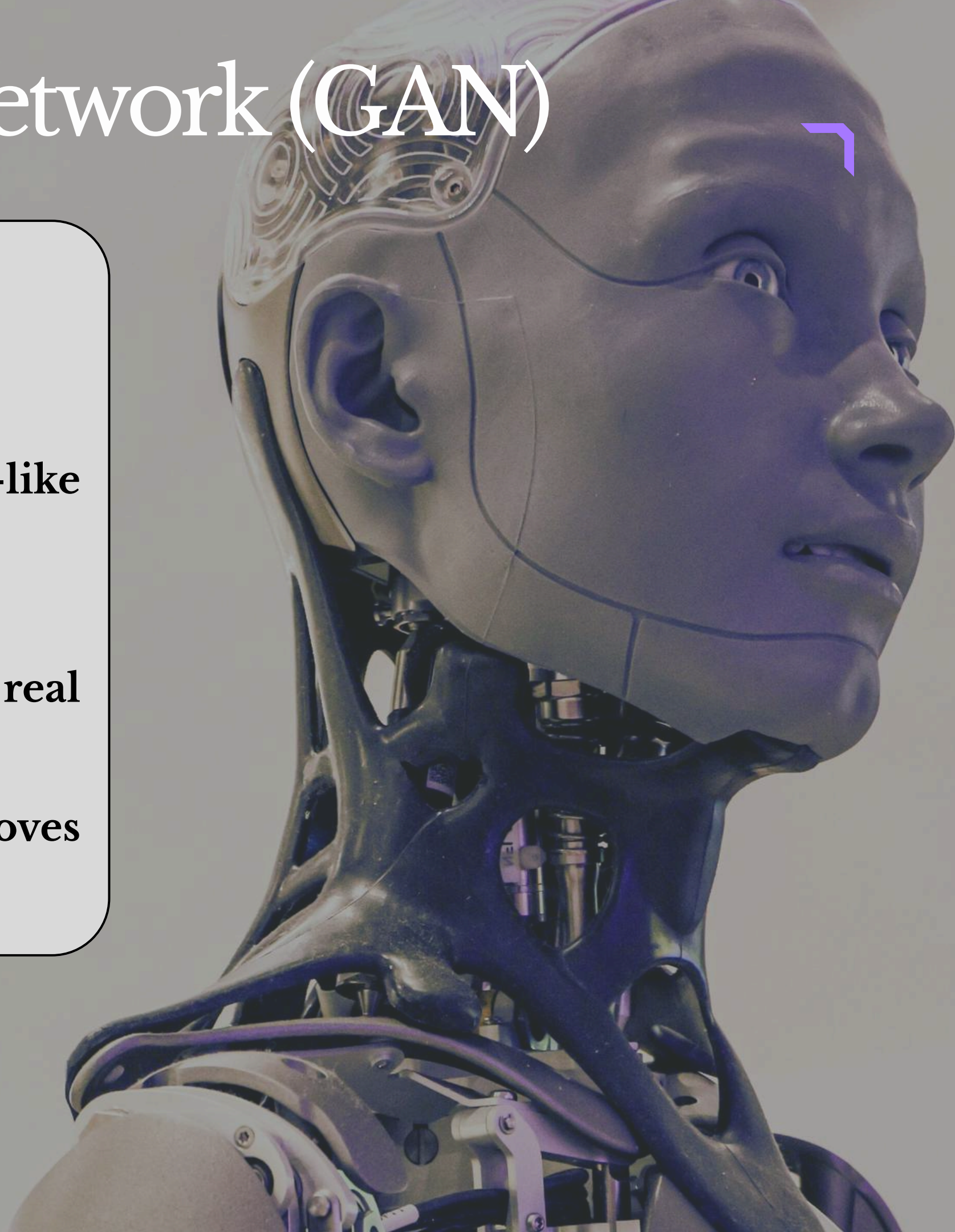
Basic Idea



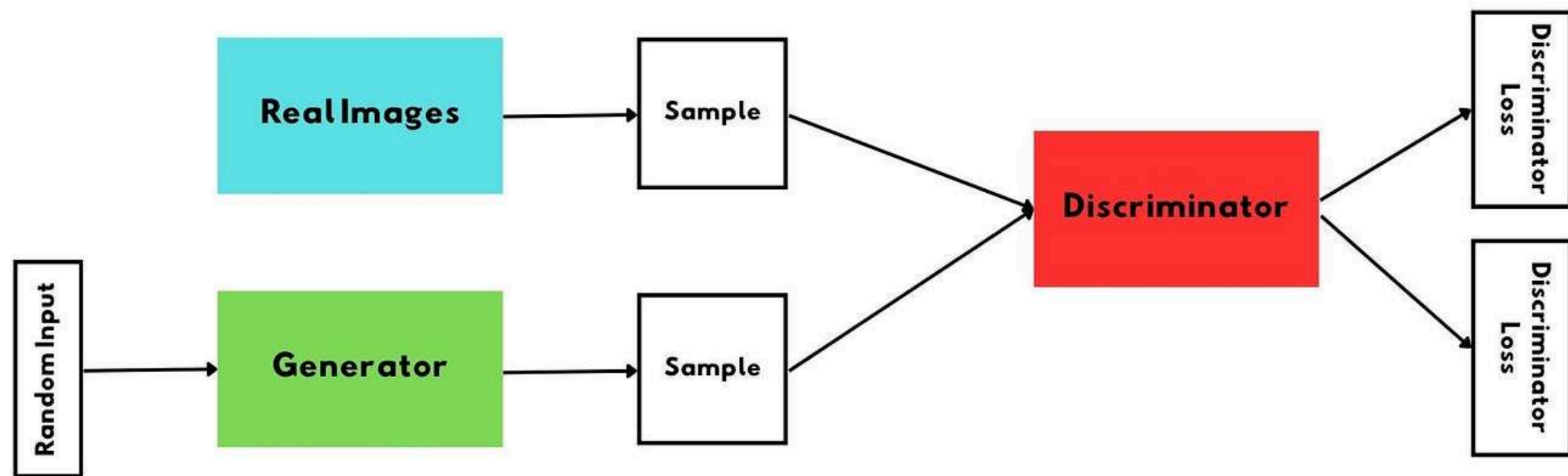
Generative Adversarial Network (GAN)

- GAN consists of two neural networks:
- Generator
- Discriminator
- Both networks compete with each other in a game-like process.
- The Generator tries to create fake data.
- The Discriminator tries to identify whether the data is real or fake.
- Through continuous competition, the Generator improves and creates highly realistic outputs.

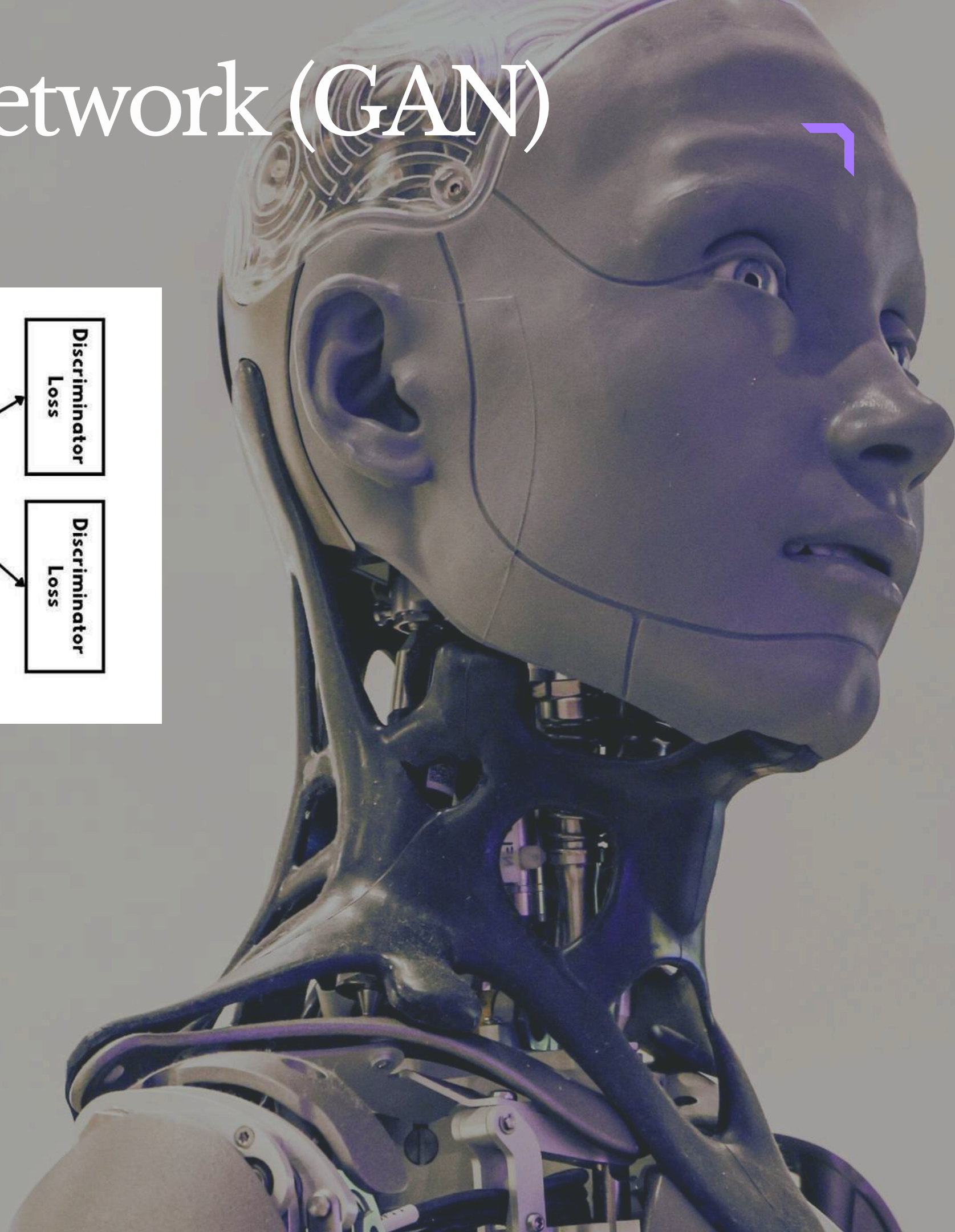
Architecture of GAN



Generative Adversarial Network (GAN)



1. Generator



Generative Adversarial Network (GAN)

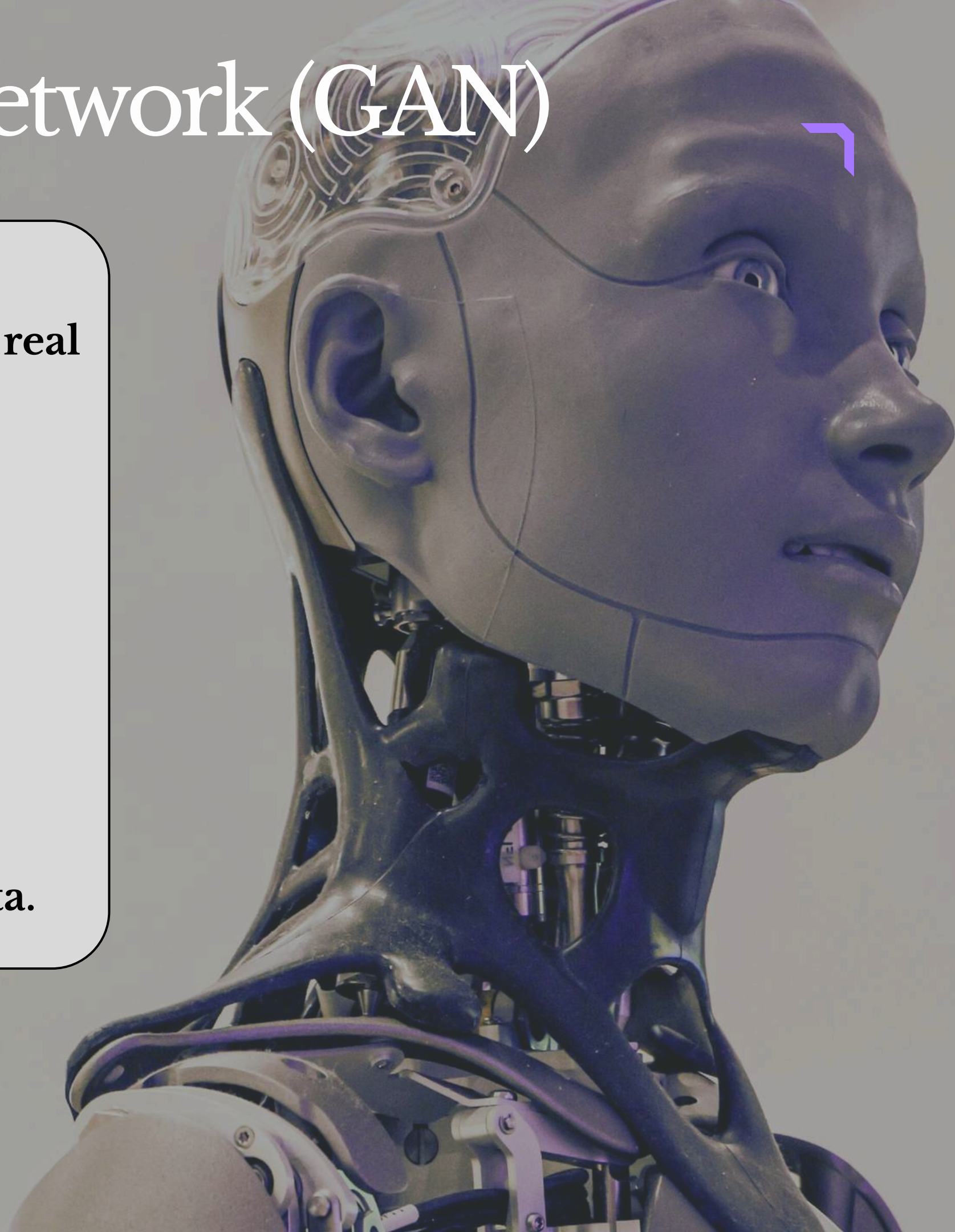
Definition

- The Generator creates fake data that looks similar to real data.

Working

- Takes random noise vector as input.
- Converts noise into synthetic images, audio, or text.
- Learns to generate realistic outputs over time.
- Goal
- Fool the Discriminator by generating realistic fake data.

2. Discriminator



Generative Adversarial Network (GAN)

Definition

- The Discriminator is a classifier that detects whether input data is real or fake.

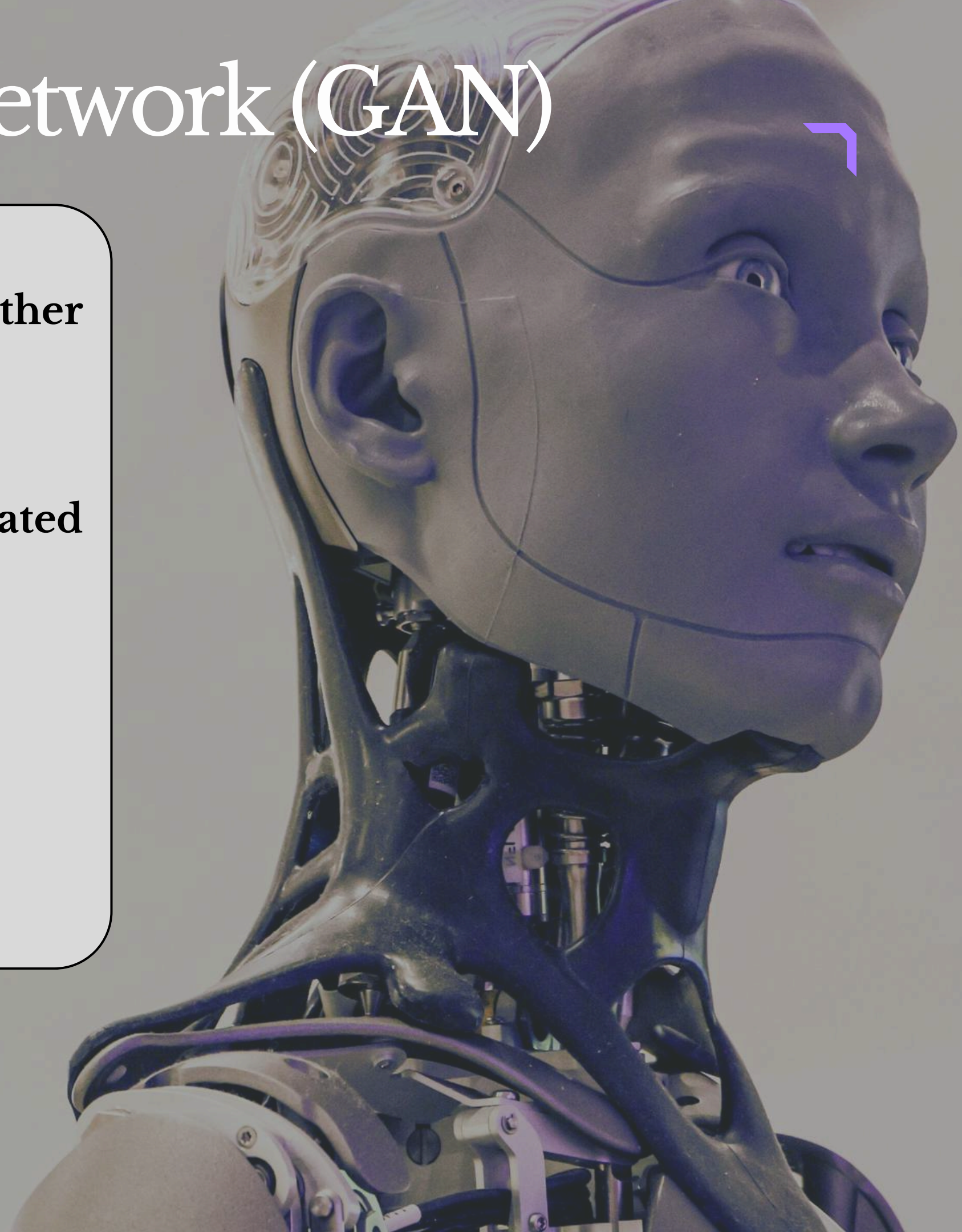
Working

- Receives both real dataset samples and fake generated samples.
- Produces probability score indicating:
 - Real Data
 - Fake Data

Goal

- Correctly identify fake and real samples.

Training Process of GAN



Generative Adversarial Network (GAN)

Step-by-Step Process

1. Generator creates fake samples using random noise.
2. Discriminator compares fake samples with real data.
3. Discriminator predicts whether data is real or fake.
4. Generator receives feedback from Discriminator.
5. Generator improves fake sample quality.
6. Discriminator also improves its detection ability.
7. Process repeats for many iterations.

Example of GAN

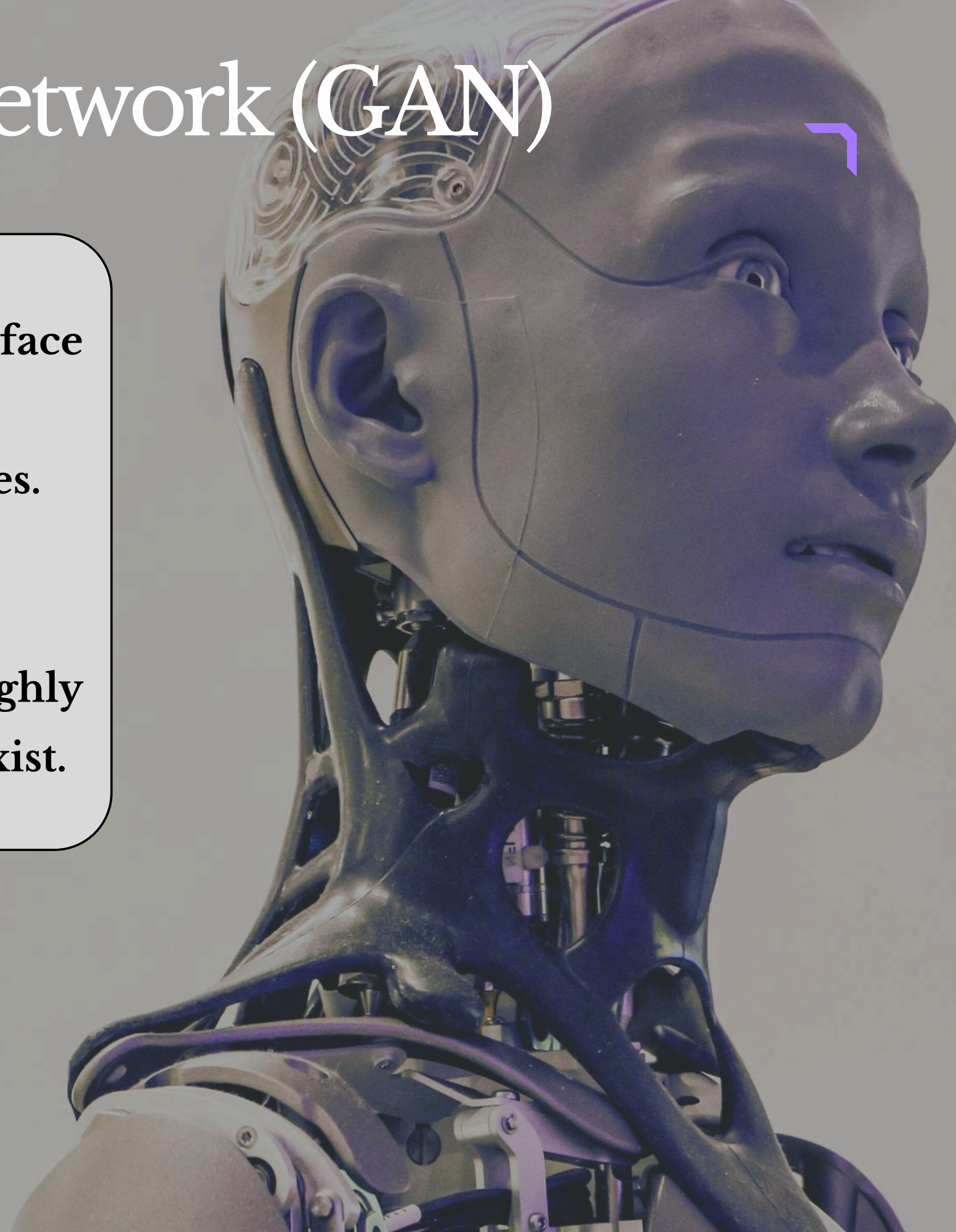


Generative Adversarial Network (GAN)

Human Face Generation

- GAN is trained using thousands of real human face images.
- Initially, Generator produces blurry or unrealistic faces.
- Discriminator easily identifies them as fake.
- Generator improves after receiving feedback.
- After many training rounds, GAN generates highly realistic human faces of people who do not actually exist.

Applications of GAN



Generative Adversarial Network (GAN)

- Image Generation
- Face Generation
- Video Generation
- Image Super Resolution
- Deepfake Technology
- Medical Imaging
- Art and Design
- Gaming and Animation



How Batch Size Affects GAN Training

Introduction

- In Generative Adversarial Networks (GANs), batch size refers to the number of training samples processed in one forward and backward pass.
- It directly affects how well the Generator and Discriminator learn together.
- GAN training is very sensitive to batch size because two networks are competing with each other.

Why Batch Size Matters in GANs?



How Batch Size Affects GAN Training

- GANs consist of a Generator and a Discriminator working in competition.
- If one network learns too fast or too slow, training becomes unstable.
- Batch size influences:
 - Learning stability
 - Gradient quality
 - Training speed
 - Quality of generated outputs

Effect of Small Batch Size



How Batch Size Affects GAN Training

Advantages

- More frequent updates to the model.
- Helps Generator and Discriminator stay more synchronized.
- Can improve adaptability during training.

Disadvantages

- Noisy gradient updates.
- Slower convergence.
- May reduce quality of generated images

Effect of Large Batch Size



How Batch Size Affects GAN Training

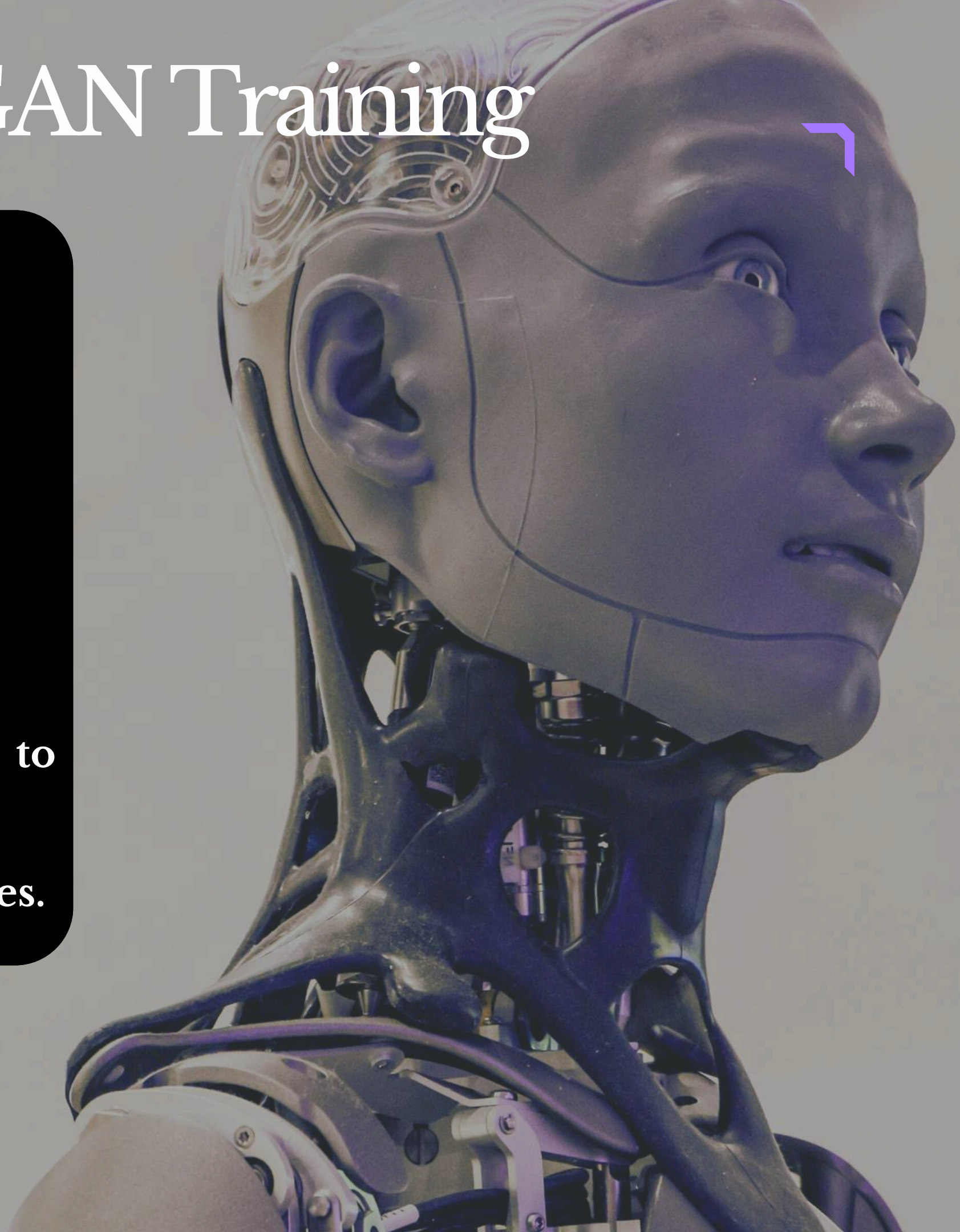
Advantages

- Produces smoother and more stable gradients.
- Often leads to better quality generated images.
- More stable training process.

Disadvantages

- Can cause overfitting.
- Discriminator may become too powerful compared to Generator.
- May lead to training instability or collapse in some cases.

Common Problems Due to
Batch Size



How Batch Size Affects GAN Training

1. Mode Collapse

- Generator produces limited types of outputs repeatedly.
- Happens when training becomes unbalanced.

2. Imbalance in Training

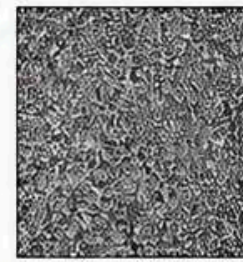
- If Discriminator learns too fast, Generator fails to improve.
- If Generator learns too fast, Discriminator becomes weak.



Applications of GAN

i) Image Generation & Image-to-Image Translation

- GANs can generate new, realistic images from random noise or modify existing images.
- This is useful in many creative fields such as art, design and game development.



Random Noise



Generated Image



Image-to-Image Translation

ii) Art Generation & Style Transfer

- GANs can learn to replicate the style of famous artists or create entirely new artworks by learning from various painting styles.
- Eg. a sketch can be transformed into a Van Gogh style painting using GAN model.
- This is known as style transfer, where the content of one image & style of another are combined into a new image.



Content Image
(Sketch)



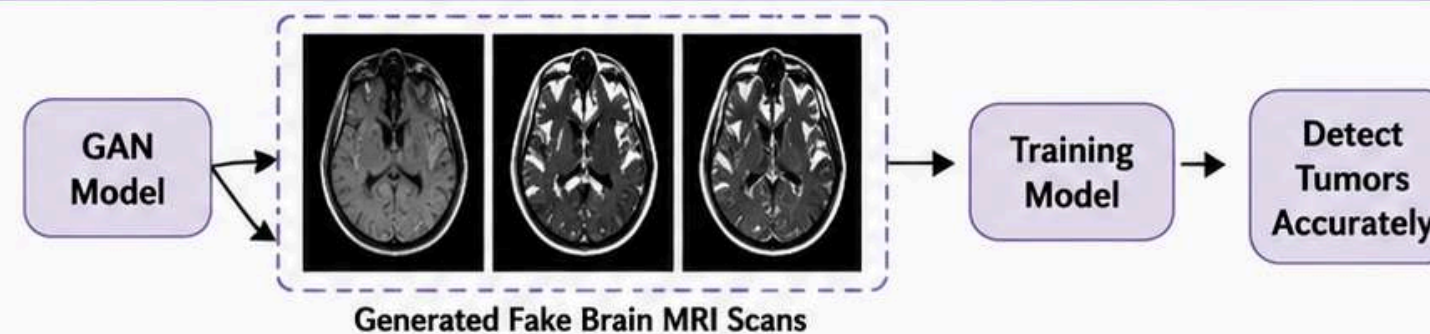
Style Image
(Van Gogh Style)



Output Image
(Style Transferred)

iii) Medical Imaging

- GANs help doctors by generating medical images like X-rays, MRI scans, especially in situations where real medical data is limited or sensitive.
- Eg, GANs can generate fake brain MRI scans to train a model to detect tumors more accurately.



iv) Deepfake Videos / Face Swapping

- GANs are used to create deepfakes – videos where a person's face is swapped / altered to mimic another person's facial expression & voice.
- Eg, an actor's face can be swapped into different scenes without re-shooting using GAN.



Source Face



Target Video



Deepfake Result

Types of GANs

1. Vanilla GAN (Basic GAN)

Definition

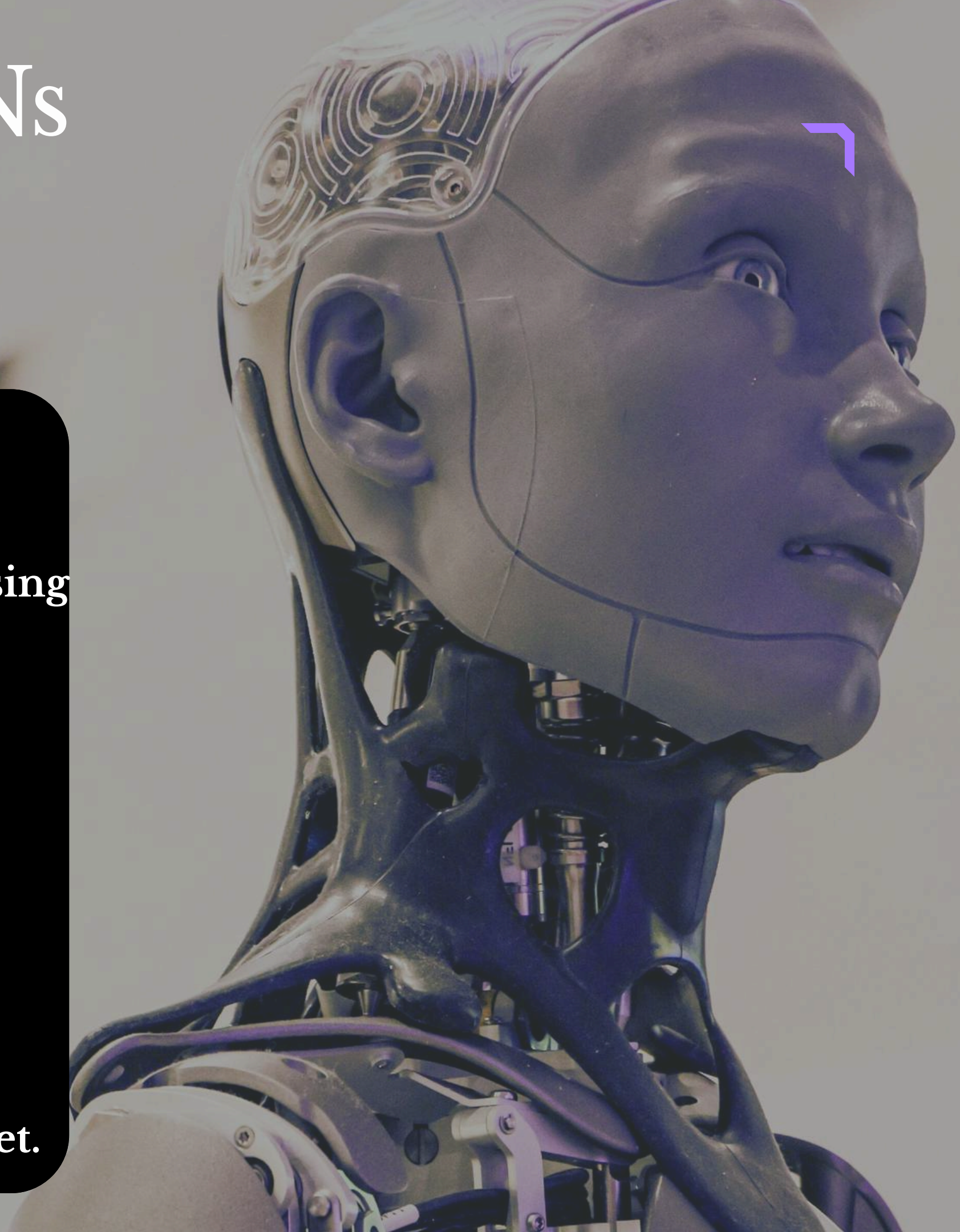
- Vanilla GAN is the original and simplest GAN model.
- It consists of a basic Generator and Discriminator using feedforward neural networks.

Working

- Generator creates fake data from random noise.
- Discriminator checks whether data is real or fake.

Example

- Generating handwritten digits similar to MNIST dataset.



Types of GANs

2. Conditional GAN (cGAN)

Definition

- cGAN is an improved GAN where generation is controlled using additional input.

Working

- Both Generator and Discriminator receive extra information (like class labels).
- Helps generate specific outputs based on condition.

Example

- Generating images of a “cat” only when the label “cat” is given.



Types of GANs

3. Deep Convolutional GAN

Definition

- DCGAN uses Convolutional Neural Networks (CNNs) instead of fully connected layers.

Features

- More powerful architecture than vanilla GAN.
- Produces high-quality and realistic images.

Example

- Generating realistic celebrity faces.



Types of GANs

4. CycleGAN

Definition

- CycleGAN is used for image-to-image translation without paired training data.

Working

- Learns mapping between two different image domains.
- Uses cycle consistency loss.

Example

- Converting photos into paintings (e.g., Van Gogh style) without paired datasets.



Types of GANs

5. Super-Resolution GAN (SRGAN)

Definition

- SRGAN improves image resolution and quality.

Working

- Generator converts low-resolution images into high-resolution images.
- Discriminator ensures generated images look realistic.

Example

- Converting 64×64 blurry images into 256×256 sharp images.



Types of GANs

6. Wasserstein GAN (WGAN)

Definition

- WGAN improves training stability using Wasserstein distance.

Features

- Uses different loss function instead of binary cross-entropy.
- Reduces training instability issues in GANs.

Example

- More stable and realistic image generation compared to vanilla GAN.

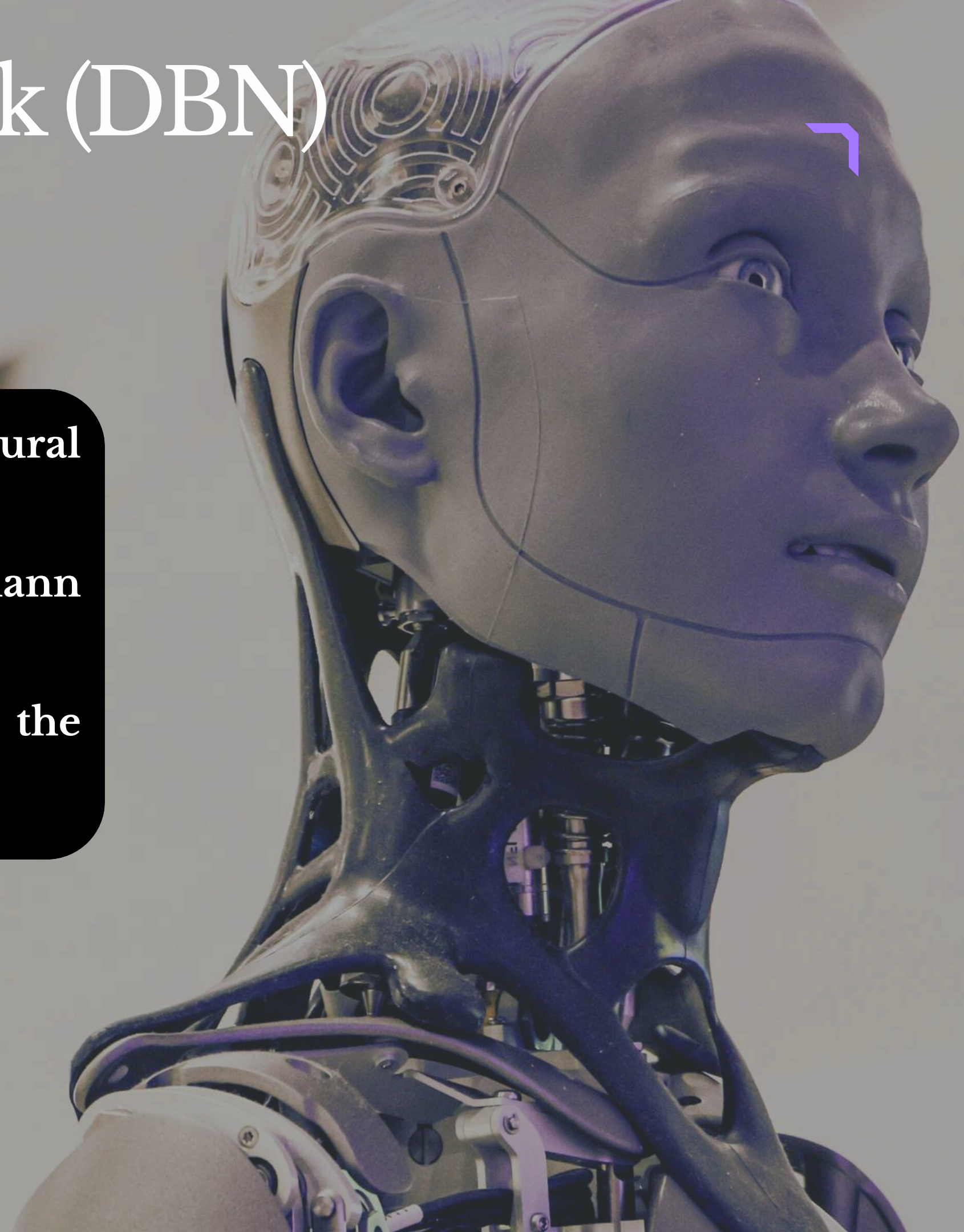


Deep Belief Network (DBN)

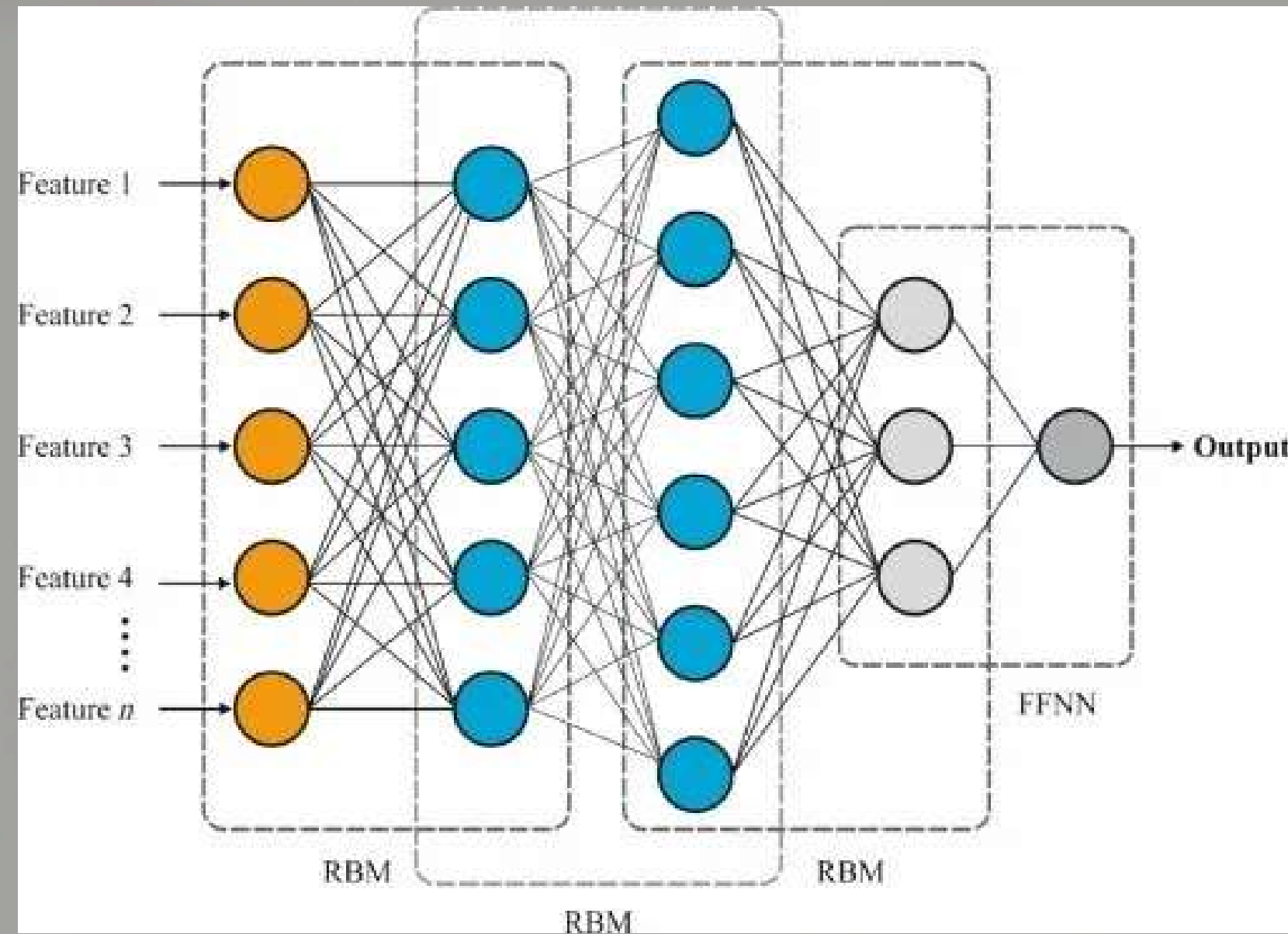
Introduction

- A Deep Belief Network (DBN) is a type of deep neural network.
- It is formed by stacking multiple Restricted Boltzmann Machines (RBMs) on top of each other.
- Each RBM learns features from the output of the previous layer.

Architecture



Deep Belief Network (DBN)



Working of DBN



Deep Belief Network (DBN)

1. Unsupervised Pre-training

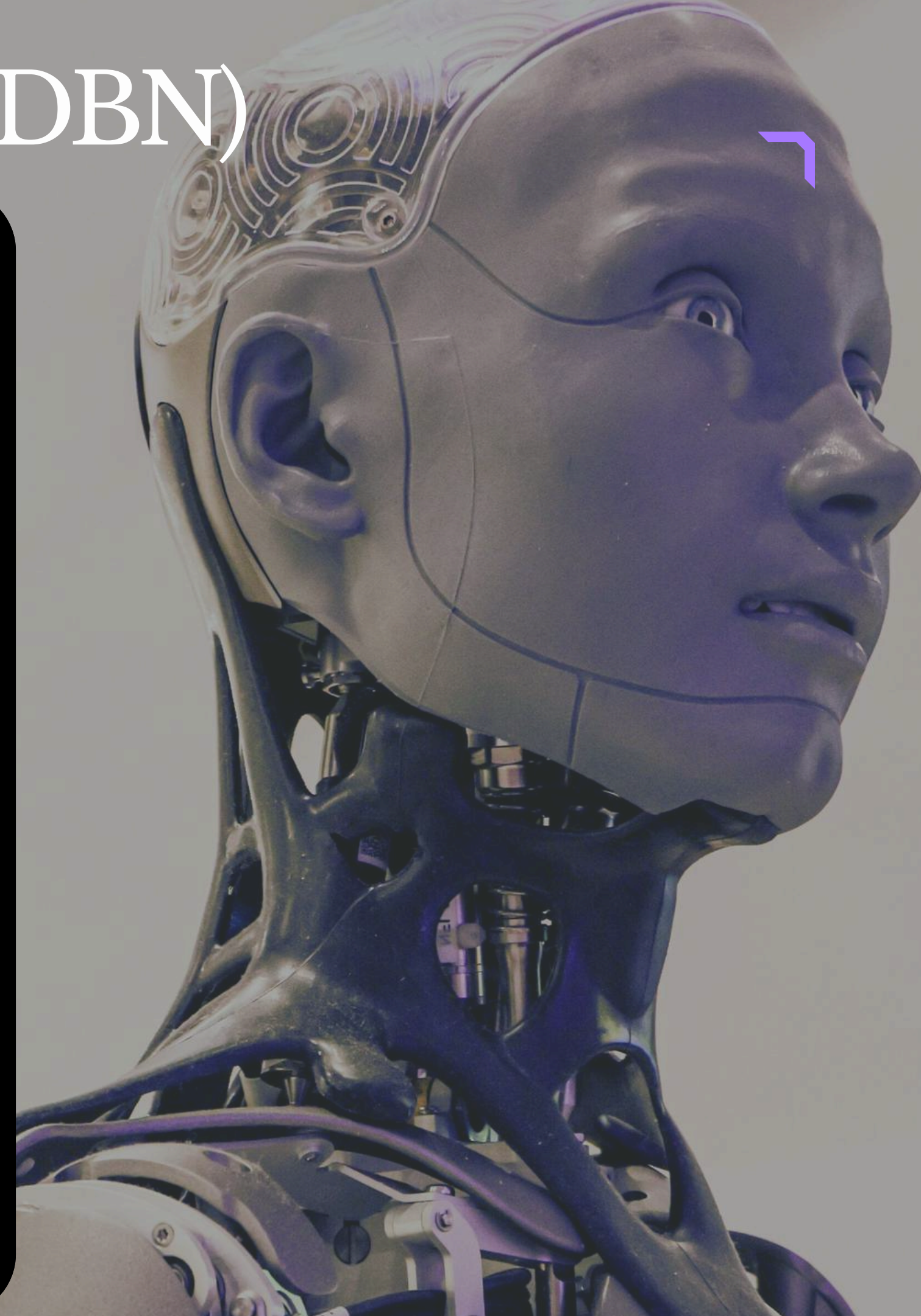
- Each RBM is trained one by one.
- The first RBM takes raw input data.
- Output of one RBM becomes input to the next RBM.
- Helps in learning useful feature representations.

2. Feature Learning

- Each layer learns different levels of features.
- Lower layers learn simple features.
- Higher layers learn complex patterns.

3. Supervised Fine-tuning

- After stacking all RBMs, the whole network is fine-tuned.
- Uses labeled data and backpropagation.

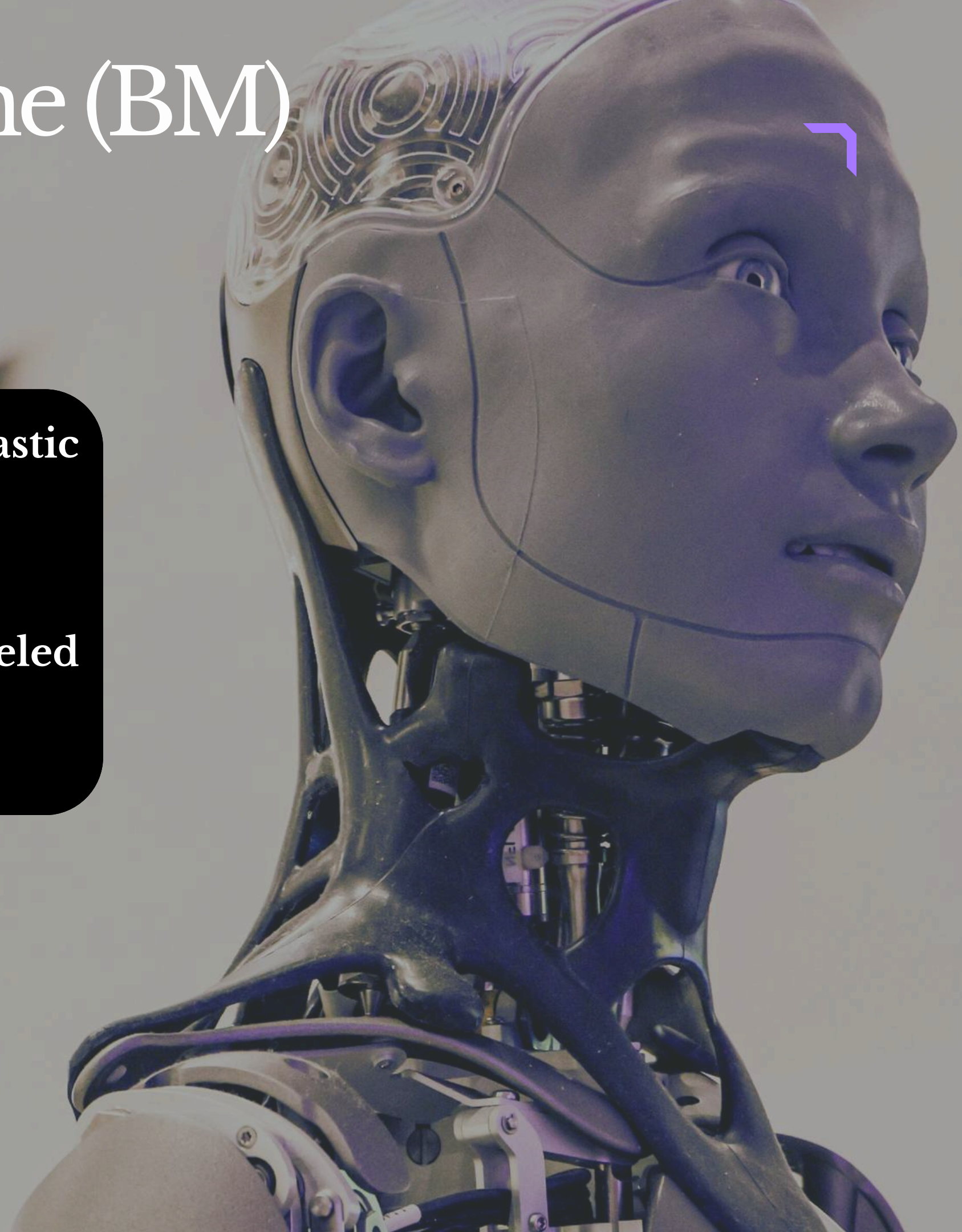


Boltzmann Machine (BM)

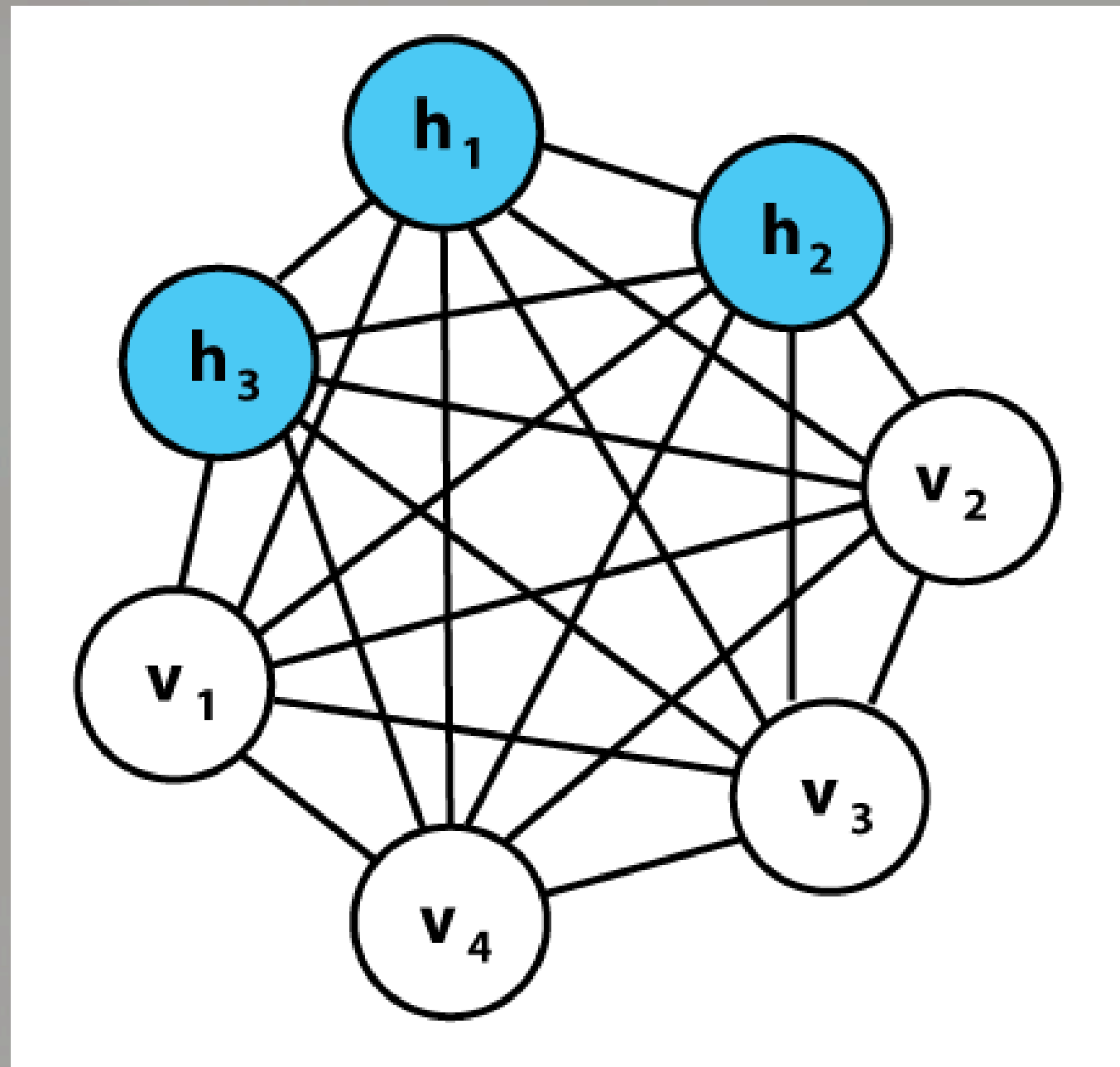
Introduction

- A Boltzmann Machine (BM) is a type of stochastic (randomized) recurrent neural network.
- It is mainly used for unsupervised learning.
- It learns patterns and features from data without labeled outputs.

Architecture



Boltzmann Machine (BM)



Working of Boltzmann Machine

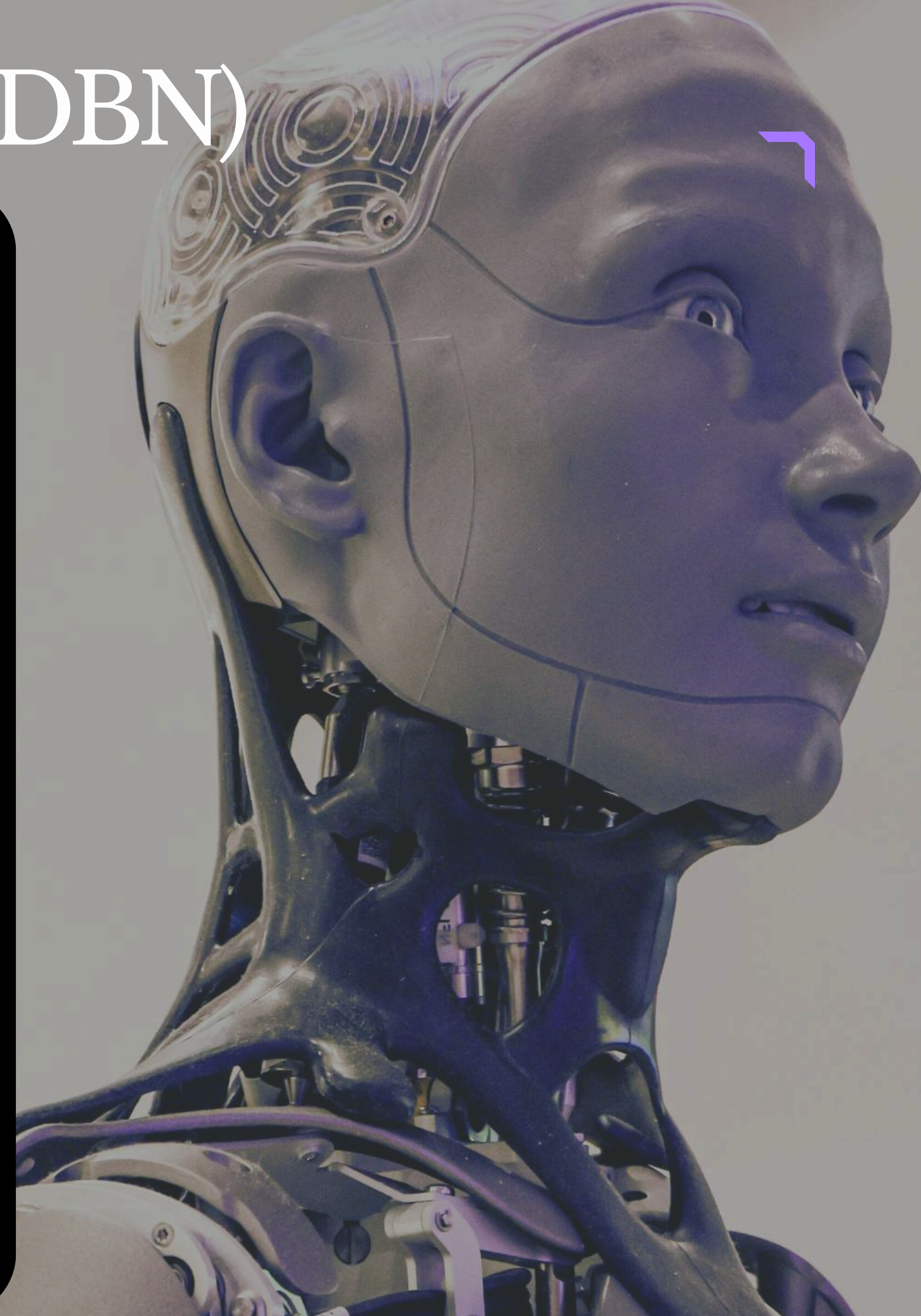


Deep Belief Network (DBN)

- All neurons are fully connected with each other.
- There are no self-connections (a neuron does not connect to itself).
- Connections are bidirectional (symmetric):
 - Weight from $A \rightarrow B$ is same as $B \rightarrow A$.
- The network updates its state dynamically to find stable patterns.

Energy Function

- Boltzmann Machine uses an energy function to measure how good a configuration is.
- Lower energy means a more likely and better representation of data.
- The network tries to minimize energy to reach an optimal state.



Deep Belief Network (DBN)

- Objectives of Boltzmann Machine
- Learn the probability distribution of data.
- Discover hidden patterns and structures without supervision.
- Minimize network energy to find stable states.
- Capture dependencies between input features.



Boltzmann Machine (BM)

Types of Boltzmann Machines

1. Standard Boltzmann Machine

- It is the original version of Boltzmann Machine.
- Every neuron is connected to every other neuron.
- Visible units are connected to other visible units.
- Hidden units are also fully interconnected.
- It can model very complex patterns and dependencies.
- However, it is computationally expensive and slow to train.

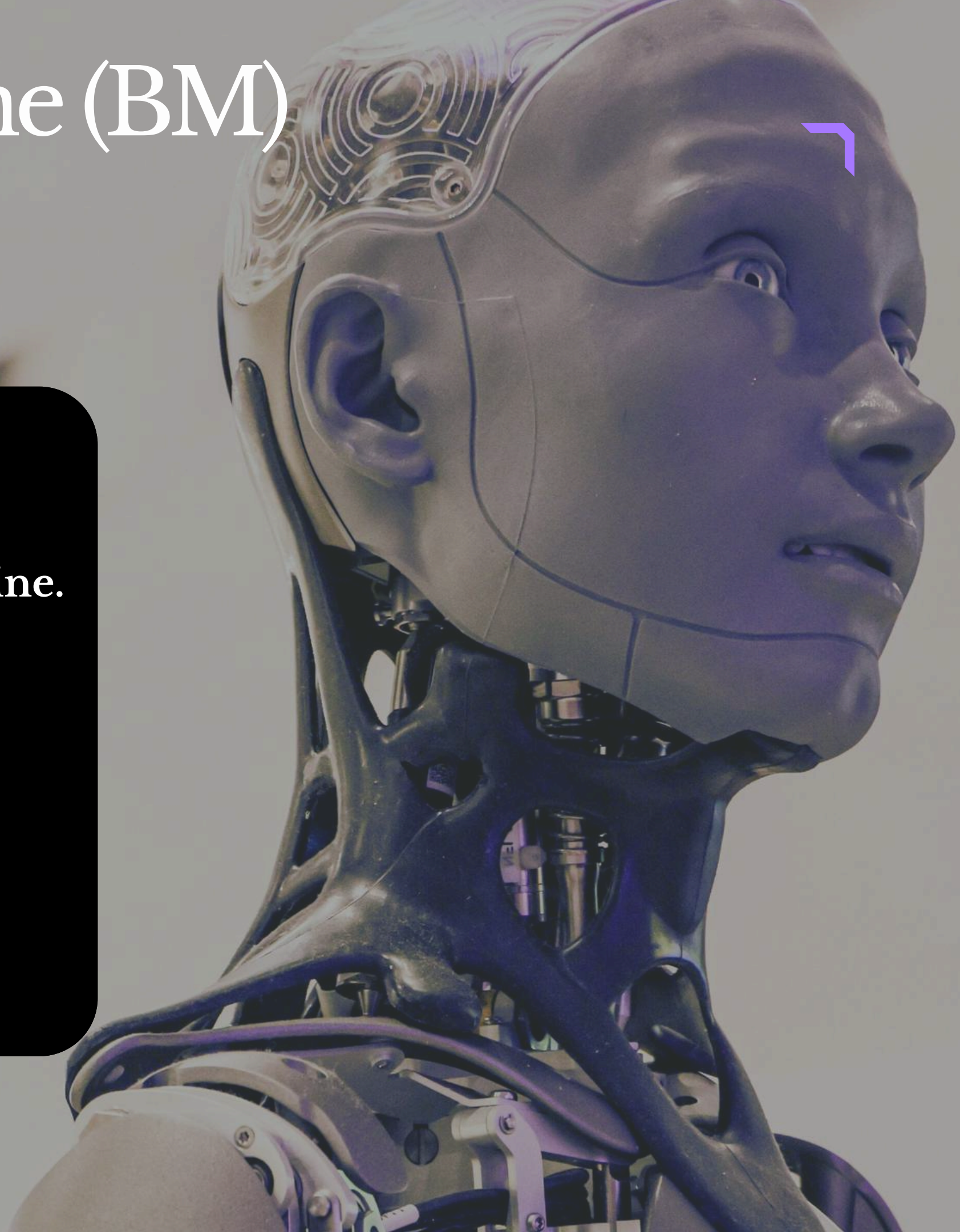


Boltzmann Machine (BM)

Types of Boltzmann Machines

2. Restricted Boltzmann Machine (RBM)

- A simplified version of the standard Boltzmann Machine.
- Connections within the same layer are removed.
- Visible units are connected only to hidden units.
- No visible-visible or hidden-hidden connections exist.
- This restriction reduces complexity significantly.
- Easier and faster to train compared to standard BM.

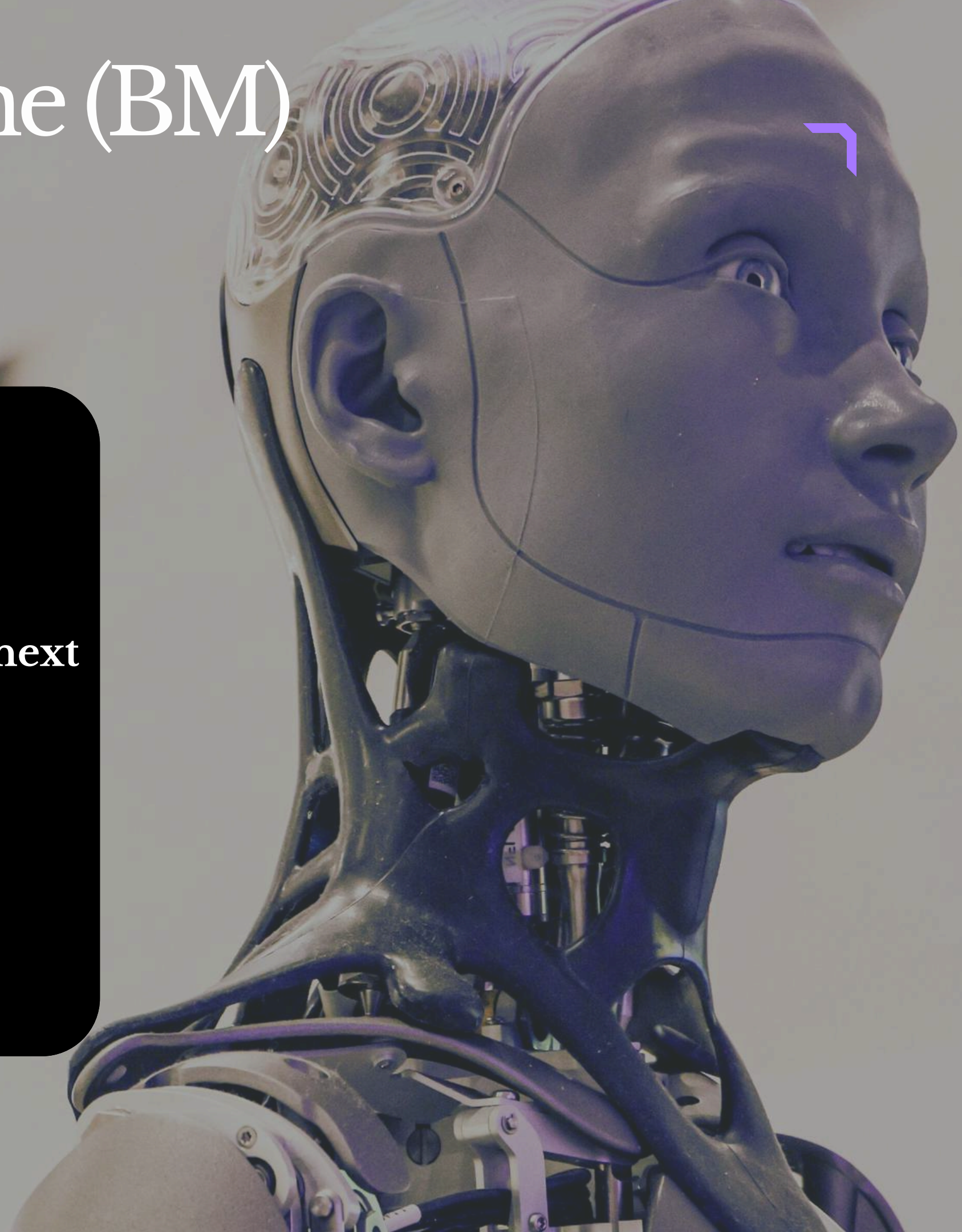


Boltzmann Machine (BM)

Types of Boltzmann Machines

3. Deep Belief Network (DBN)

- A DBN is formed by stacking multiple RBMs.
- The output of one RBM becomes the input to the next RBM.
- Helps in learning hierarchical feature representations.
- Learns from low-level features to high-level features.
- Used for deep unsupervised feature learning.



Difference between Generative & Descriptive model

Generative Model	Discriminative Model
1. Models the joint probability $P(X, Y)$	1. Models the conditional probability $P(Y$
2. Learns how data is generated	2. Learns decision boundary between classes
3. Can generate new data samples	3. Cannot generate new data
4. Focuses on understanding data distribution	4. Focuses only on classification
5. Works by modeling each class separately	5. Directly separates classes
6. Examples: Naive Bayes, GAN, RBM	6. Examples: Logistic Regression, SVM, CNN
7. Generally more flexible but complex	7. Usually more accurate for prediction tasks





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